IS101 — Introduction to Computing

Inventory & Budget Workbook

Oasis Events · Unit 3 — Spreadsheets · Variation 2 · 100 points · 25 tasks

How to use this document: Download the starter workbook (Excel_Var2_Starter.xlsx) from the Project Grader page. Open it in Microsoft Excel 365 and keep this sheet open beside it. Work through each task in order. The Inventory sheet has 8 items (rows 4–11) with columns for Item ID, Item Name, Category, Qty, Unit Cost, Line Total, SKU, Category Code, and Item Label. The PriceList sheet tab has item names and unit costs for lookups. When you're done, save the file and upload it to the Variation 2 grader for instant scoring. **Important:** When writing formulas, always reference cell addresses, not column name words.

Workbook Setup Tasks 1–4

Task 1 2 points

Turn on Show Formulas

WHY THIS MATTERS

Show Formulas mode displays every formula in its cell instead of the result — useful for auditing.

WHERE TO FIND IT

Formulas tab → Formula Auditing group → Show Formulas (or press Ctrl+`).

HOW YOU'LL KNOW IT WORKED

Every cell shows its formula text instead of calculated values. Column widths widen automatically.

Task 2 3 points

Set workbook properties: Title and Author

WHY THIS MATTERS

Metadata makes the workbook identifiable and searchable.

WHERE TO FIND IT

File tab → Info → Properties panel. Set Title to “Event Inventory & Budget” and Author to your name.

HOW YOU'LL KNOW IT WORKED

File {ARR} Info shows Title and Author populated.

Task 3 3 points

Use SEQUENCE to generate the line-item ID column

WHY THIS MATTERS

SEQUENCE creates a series of sequential numbers automatically. Use it to fill the Item ID column (A).

WHERE TO FIND IT

Click cell A4 → type a SEQUENCE formula that generates 8 numbers (one for each item, rows 4–11) → Enter.

HOW YOU'LL KNOW IT WORKED

Column A shows 1 through 8 in cells A4:A11, generated by a single formula.

Task 4 4 points

Format Unit Cost and Line Total columns as Currency

WHY THIS MATTERS

Currency formatting adds dollar signs and two decimal places to financial data.

WHERE TO FIND IT

Select the Unit Cost column data (E4:E11) and the Line Total column data (F4:F11) → Home tab → Number group → Currency.

HOW YOU'LL KNOW IT WORKED
Values show as $12.00 format with dollar signs and two decimals.

Tables & Data Tasks 5–9

Task 5 3 points

Convert the inventory data to an Excel Table

WHY THIS MATTERS
Excel Tables add automatic filtering, banded rows, and structured references.

WHERE TO FIND IT
Click anywhere in the inventory data → use any method: Ctrl+T, Insert tab → Table, or Home tab → Format as Table → verify the range → OK.

HOW YOU'LL KNOW IT WORKED
Filter arrows appear on each column header; the Table Design tab appears when you click in the table.

Task 6 4 points

Add a Line Total column to the table

WHY THIS MATTERS
If the Line Total column (F) is not already inside the table range, you need to extend the table to include it.

WHERE TO FIND IT
Click in the table → Table Design tab → Resize Table → adjust the range to include the Line Total column (F) → OK. Or simply click in a cell in column F within the table area and start typing a formula — Excel auto-expands the table.

HOW YOU'LL KNOW IT WORKED
The Line Total column (F) is inside the table range, shown by the table formatting extending to column F.

Task 7 4 points

Add a Total Row summing the Line Total column

WHY THIS MATTERS
The Total Row must be enabled through the Table Design tab, not by typing a manual SUM formula.

WHERE TO FIND IT
Click anywhere in the table → Table Design tab → check the Total Row checkbox. Then click the Total Row cell under the Line Total column (F) → click the dropdown arrow → select Sum.

HOW YOU'LL KNOW IT WORKED
A Total Row appears at the bottom of the table. The cell under Line Total shows the sum.

Task 8 6 points

Use a structured reference for the Line Total

WHY THIS MATTERS
Structured references use table column names instead of cell addresses. In the Line Total column (F), multiply Qty by Unit Cost using the @ symbol to reference the current row.

WHERE TO FIND IT
Click cell F4 → type a formula that multiplies the current row's Qty by the current row's Unit Cost using structured references with [@column name] syntax → Enter. The formula should auto-fill down for all rows.

HOW YOU'LL KNOW IT WORKED
The formula bar shows a structured reference with [@...] syntax, not regular cell addresses like D4*E4.

Task 9 6 points

Use SUMIF to total spend per category

WHY THIS MATTERS

SUMIF adds values that match a condition. Use it to calculate total spend for each category. The summary area is in cells K8:L14, with category names in column K and spend totals in column L.

WHERE TO FIND IT

Click cell L9 → type a SUMIF formula that searches the Category column (C4:C11) for the category name in column K, and sums the corresponding Line Total values (F4:F11) → Enter. Copy down for each category (L9:L14).

HOW YOU'LL KNOW IT WORKED

Each category in column K shows its total spend in column L.

Text & Lookup Formulas Tasks 10–16

Task 10 4 points

Use LEFT, RIGHT, or MID to build a short SKU

WHY THIS MATTERS

Text functions extract parts of a string. Build a SKU code in column G by combining characters from the Item Name and Category.

WHERE TO FIND IT

Click cell G4 → type a formula using UPPER with LEFT and/or MID to extract characters from the Item Name (B4) and Category (C4) → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Column G shows short codes like “FOL-Fu” for each item.

Task 11 4 points

Use UPPER and LEN for a category code

WHY THIS MATTERS

UPPER converts text to all capitals; LEN counts characters. Combine them to create a category code in column H.

WHERE TO FIND IT

Click cell H4 → type a formula that combines UPPER of the Category (C4) with the LEN of the Category → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Column H shows codes like “FURNITURE9” — the uppercase category name followed by its character count.

Task 12 5 points

Use TEXTJOIN to build a combined item label

WHY THIS MATTERS

TEXTJOIN concatenates multiple values with a delimiter between them. Build a label in column I combining Item Name, Category, and Qty.

WHERE TO FIND IT

Click cell I4 → type a TEXTJOIN formula using a delimiter like “ | ” (pipe with spaces), skipping empty cells, and joining the Item Name (B4), Category (C4), and Qty (D4) → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Column I shows labels like “Folding Chair | Furniture | 120” for each item.

Task 13 6 points

Use XLOOKUP to return each item’s unit cost from the PriceList

WHY THIS MATTERS

XLOOKUP searches for a value and returns a corresponding result from another column. Look up each Item Name (column B) on the PriceList sheet to fill in the Unit Cost (column E).

WHERE TO FIND IT

Click cell E4 → type an XLOOKUP formula that searches for the Item Name (B4) in the PriceList sheet's item names and returns the matching unit cost. Use absolute references ($) on the PriceList ranges so the formula copies correctly → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Column E shows the correct unit cost for each item, pulled from the PriceList sheet.

Task 14 4 points

Use SORT() to produce a sorted vendor list

WHY THIS MATTERS

The SORT function creates a dynamically sorted list. Place it in the Sorted Vendors area (column O).

WHERE TO FIND IT

Click cell O4 → type a SORT formula referencing the Vendor column (M4:M8) → Enter.

HOW YOU'LL KNOW IT WORKED

Column O shows the vendor names sorted alphabetically. The list updates automatically if vendors change.

Task 15 4 points

Use UNIQUE() to list distinct categories

WHY THIS MATTERS

The UNIQUE function removes duplicates from a list. Place it in the Categories area (column P).

WHERE TO FIND IT

Click cell P4 → type a UNIQUE formula referencing the Category column (C4:C11) → Enter.

HOW YOU'LL KNOW IT WORKED

Column P shows each category name once, with no duplicates.

Task 16 5 points

Use YEAR, MONTH, and DAY to break out the event date

WHY THIS MATTERS

Date functions extract parts of a date. The event date is in cell K4. Break it into Year, Month, and Day in columns R, S, and T.

WHERE TO FIND IT

Click cell R4 → type =YEAR(K4) → Enter. Click S4 → type =MONTH(K4) → Enter. Click T4 → type =DAY(K4) → Enter.

HOW YOU'LL KNOW IT WORKED

R4 shows 2026, S4 shows 9, T4 shows 15.

Conditional Formatting & Named Range Tasks 17–18

Task 17 5 points

Apply an icon set to flag items at reorder level

WHY THIS MATTERS

Icon sets add visual indicators (arrows, flags, or traffic lights) based on cell values.

WHERE TO FIND IT

Select the Qty column data (D4:D11) → Home tab → Conditional Formatting → Icon Sets → choose a style (e.g., 3 Arrows or 3 Flags).

HOW YOU'LL KNOW IT WORKED
Each cell in the Qty column shows a colored icon indicating the relative quantity level.

Task 18 4 points

Name the price-list range “PriceList”

WHY THIS MATTERS
Named ranges make formulas easier to read.

WHERE TO FIND IT
Click the PriceList sheet tab → select the price data range (A4:B11) → click in the Name Box (left of the formula bar) → type PriceList → press Enter.

HOW YOU'LL KNOW IT WORKED
The Name Box shows “PriceList” when the range is selected.

Charts Tasks 19–21

Task 19 4 points

Create a Pie chart of spend by category

WHY THIS MATTERS
A Pie chart shows how total spend is divided among categories.

WHERE TO FIND IT
Select the category names and spend totals in the SUMIF summary area (K9:L14) → Insert tab → Charts group → Pie → choose a style.

HOW YOU'LL KNOW IT WORKED
A pie chart appears showing each category’s share of total spend.

Task 20 4 points

Apply a chart style

WHY THIS MATTERS
Chart styles change the overall look of the chart — colors, effects, and layout.

WHERE TO FIND IT
Click the chart → Chart Design tab → Chart Styles gallery → choose a style.

HOW YOU'LL KNOW IT WORKED
The chart’s appearance changes to the selected style.

Task 21 3 points

Add alt text to the chart

WHY THIS MATTERS
Alt text describes the chart for screen readers and accessibility.

WHERE TO FIND IT
Right-click the chart → Edit Alt Text → type a description of what the chart shows.

HOW YOU'LL KNOW IT WORKED
Right-clicking the chart and selecting Edit Alt Text shows your description.

Finishing Touches Tasks 22–25

Task 22 4 points

Add a header with the company name and a footer with page number

WHY THIS MATTERS

Headers and footers identify printed pages. **Important:** Use the Page Setup method below, not Insert {ARR} Header & Footer, which switches views and can unfreeze your panes from Task 24.

WHERE TO FIND IT

Page Layout tab → click the Page Setup dialog launcher (small arrow in the bottom-right corner of the Page Setup group) → Header/Footer tab → Custom Header: type “Oasis Events” → Custom Footer: insert page number → OK.

HOW YOU'LL KNOW IT WORKED

Print Preview shows the company name at the top and page number at the bottom.

Task 23 3 points

Merge and center the title banner

WHY THIS MATTERS

Merging creates a wide banner that spans the data columns.

WHERE TO FIND IT

Select cells A1 through the last data column in row 1 → Home tab → Alignment group → Merge & Center.

HOW YOU'LL KNOW IT WORKED

The title “Oasis Events — Inventory & Budget” spans all data columns in one centered cell.

Task 24 3 points

Freeze the header row

WHY THIS MATTERS

Freezing keeps the column headers visible while scrolling down through inventory items.

WHERE TO FIND IT

Click cell A4 (the first data row) → View tab → Freeze Panes → Freeze Panes.

HOW YOU'LL KNOW IT WORKED

When you scroll down, row 3 (the headers) stays visible at the top.

Task 25 3 points

Sort by Category then Item Name

WHY THIS MATTERS

Multi-level sorting groups items by category and alphabetizes within each group.

WHERE TO FIND IT

Data tab → Sort → Sort by: Category (column C) → A to Z → Add Level → Then by: Item Name (column B) → A to Z → OK.

HOW YOU'LL KNOW IT WORKED

Items are grouped by category; within each category, items are alphabetical.

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